

Layla's Shape Lab

Area & Perimeter Practice · Parallelograms · Triangles · Trapezoids · Weird Shapes

Date: _____

 Perimeter = fence AROUND → units (ft)  Area = carpet INSIDE → square units (ft²)  height = dashed $\perp$

A. Area or Perimeter? Circle A or P.

 Fence around a garden **A** **P**

 Paint a wall **A** **P**

 Ribbon around a frame **A** **P**

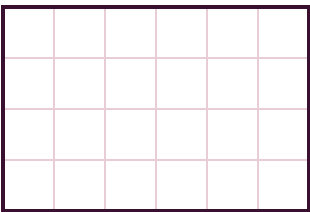
 Tiles on a floor **A** **P**

 An answer of 36 ft² **A** **P**

 An answer of 22 m **A** **P**

B. Warm-up Each square = 1 unit.

7 Count it



A = _____ units² P = _____ units

 Write it every time

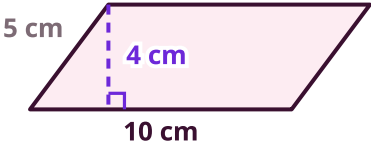
1) formula → 2) plug in numbers → 3) answer + unit

Example: $A = b \times h = 8 \times 3 = 24 \text{ cm}^2$

Slanted side? **Perimeter only.**

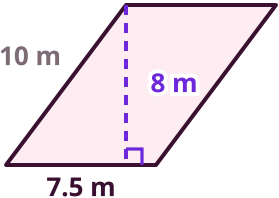
C. Parallelograms $A = b \times h$

8 Find A and P



A = _____ P = _____

9 Find A

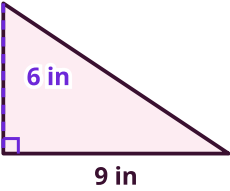


A = _____

D. Triangles $A = \frac{1}{2} \times b \times h$

10

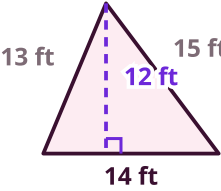
Right



A = _____

11

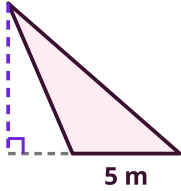
Find A and P



A = _____ P = _____

12

Height outside!

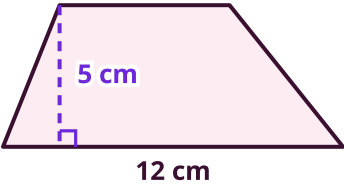


A = _____

E. Trapezoids $A = \frac{1}{2} \times (b_1 + b_2) \times h$

13

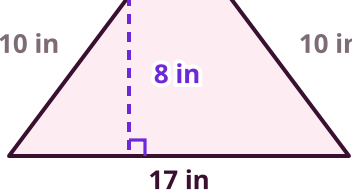
Find A



A = _____

14

Find A and P

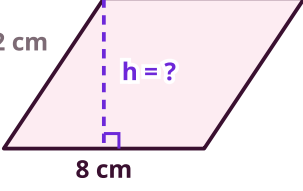


A = _____ P = _____

F. Backwards Given the area, find the missing part.

15

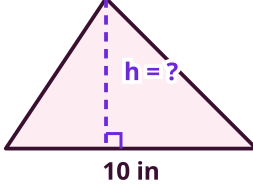
$A = 48 \text{ cm}^2$



h = _____

16

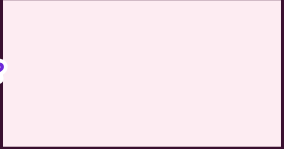
$A = 30 \text{ in}^2$



h = _____

17

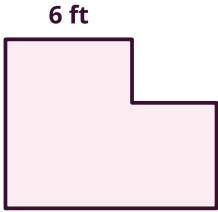
Rectangle: $P = 26 \text{ ft}$



w = _____

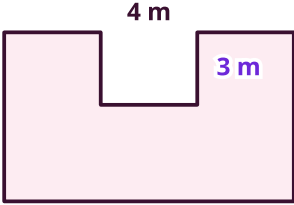
G. Weird shapes Split it + or cut it out -

18Find A and P



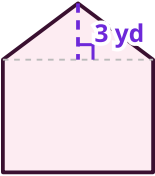
A = _____ P = _____

19Big – cut-out



A = _____

20Rectangle + triangle



A = _____

H. Word problems Draw it. Label it. Answer + unit!

21

A garden is **12.5 ft** long and **6 ft** wide.

a) How much fence goes around it?

b) How much soil covers it?

a) _____ b) _____

22

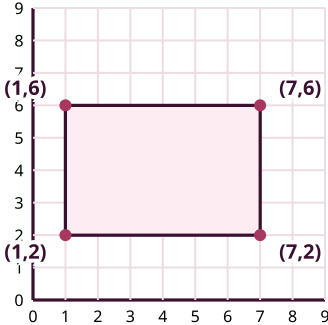
A triangle pennant has a base of **10 in** and a height of **16 in**. How much felt is needed?

23

A window is a trapezoid. Bases: **9 ft** and **5 ft**. Height: **4 ft**. How much glass?

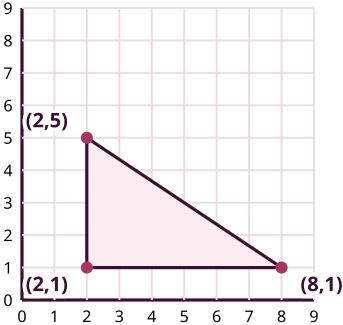
I. Coordinate plane

24Rectangle



A = _____ P = _____

25Right triangle



A = _____

🦋 Layla's Cheat Sheet 🦋

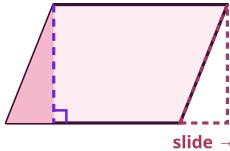
Area & Perimeter — keep this next to you!



Fence vs Carpet

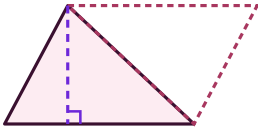
Perimeter = walk AROUND, add every outside side → **ft**

Area = cover INSIDE → **ft²** (little ²)



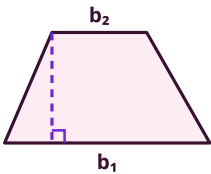
Parallelogram: $A = b \times h$

"Slide the slice" — cut the triangle off, slide it over → a rectangle.



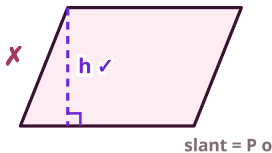
Triangle: $A = \frac{1}{2} \times b \times h$

"A triangle is half a parallelogram." Or $b \times h \div 2$.



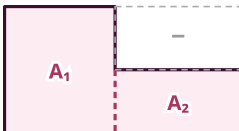
Trapezoid: $A = \frac{1}{2} \times (b_1 + b_2) \times h$

"Add the parallels, times height, cut in half." (2 copies = 1 parallelogram)



Height stands up straight

Use the dashed height with the square corner. Slanted side → **perimeter only**. Height can be **outside** an obtuse triangle.



Weird shape? Split **+** or cut **-**

Split into rectangles/triangles and add, OR big rectangle – the missing piece. Find missing sides: opposite sides match.

⚠ Watch out: slanted side $\neq$ height · don't forget the $\frac{1}{2}$ · ft vs ft² · height is never part of the perimeter · count every side on L-shapes (none twice)

Backwards: $A = b \times h \rightarrow h = A \div b$. Triangle: $h = A \times 2 \div b$. **Coordinates:** length = big – small. **Always:** formula → numbers → answer + unit 

Answer Key — Layla's Shape Lab

Check AFTER you try. Look at the work, not just the answer.

#	Answer	Work
A1	P	fence goes around
A2	A	covers the wall
A3	P	goes around the edge
A4	A	covers the floor
A5	A	square units → area
A6	P	plain units → perimeter
7	A = 24 units²; P = 20 units	$A = 6 \times 4 = 24$; $P = 6+4+6+4 = 20$
8	A = 40 cm²; P = 30 cm	$A = b \times h = 10 \times 4 = 40 \text{ cm}^2$ (not $\times 5$). $P = 10+5+10+5 = 30 \text{ cm}$ (not 4)
9	A = 60 m²	$A = 7.5 \times 8 = 60 \text{ m}^2$ (10 m slant not used)
10	A = 27 in²	$\frac{1}{2} \times 9 \times 6 = 54 \div 2 = 27 \text{ in}^2$
11	A = 84 ft²; P = 42 ft	$A = \frac{1}{2} \times 14 \times 12 = 84 \text{ ft}^2$ (13 & 15 not used). $P = 14+13+15 = 42 \text{ ft}$ (12 not used)
12	A = 17.5 m²	$\frac{1}{2} \times 5 \times 7 = 35 \div 2 = 17.5 \text{ m}^2$ (height is outside, still use it)
13	A = 45 cm²	$\frac{1}{2} \times (12+6) \times 5 = \frac{1}{2} \times 18 \times 5 = 45 \text{ cm}^2$
14	A = 88 in²; P = 42 in	$A = \frac{1}{2} \times (17+5) \times 8 = \frac{1}{2} \times 22 \times 8 = 88 \text{ in}^2$. $P = 17+5+10+10 = 42 \text{ in}$
15	h = 6 cm	$48 = 8 \times h \rightarrow h = 48 \div 8 = 6 \text{ cm}$
16	h = 6 in	$30 = \frac{1}{2} \times 10 \times h \rightarrow 30 = 5 \times h \rightarrow h = 6 \text{ in}$
17	w = 5 ft	$26 - 8 - 8 = 10 \rightarrow w = 10 \div 2 = 5 \text{ ft}$
18	A = 68 ft²; P = 36 ft	Missing sides: $8-5 = 3 \text{ ft}$, $10-6 = 4 \text{ ft}$. $A = 6 \times 8 + 4 \times 5 = 48 + 20 = 68 \text{ ft}^2$ (or $10 \times 8 - 4 \times 3 = 68$). $P = 6+3+4+5+10+8 = 36 \text{ ft}$
19	A = 72 m²	$12 \times 7 - 4 \times 3 = 84 - 12 = 72 \text{ m}^2$
20	A = 60 yd²	$8 \times 6 + \frac{1}{2} \times 8 \times 3 = 48 + 12 = 60 \text{ yd}^2$
21	a) 37 ft b) 75 ft²	a) $P = 12.5+6+12.5+6 = 37 \text{ ft}$; b) $A = 12.5 \times 6 = 75 \text{ ft}^2$
22	80 in²	$\frac{1}{2} \times 10 \times 16 = 80 \text{ in}^2$
23	28 ft²	$\frac{1}{2} \times (9+5) \times 4 = \frac{1}{2} \times 14 \times 4 = 28 \text{ ft}^2$
24	A = 24 units²; P = 20 units	length $7-1 = 6$, height $6-2 = 4 \rightarrow A = 24$, $P = 6+4+6+4 = 20$
25	A = 12 units²	base $8-2 = 6$, height $5-1 = 4 \rightarrow \frac{1}{2} \times 6 \times 4 = 12$